

BIRZEIT UNIVERSITY

Faculty Of Engineering And Technology

Computer Science Department

DATA BASE SYSTEMS COMP333

Project Report

Dubai Jewelry: Online Store & Dynamic Pricing Database

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1. Project Description:

The Dubai Jewelry Management System is a specialized platform designed to manage the sale of gold ornaments based on real-time market fluctuations. The system organizes products into different gold purity levels (21K, 18K, and 14K), catering to customers who seek transparency in pricing and quality. Customers can easily browse through various categories such as necklaces, bracelets, and rings, viewing up-to-date prices that reflect the current global gold rate. By centralizing inventory and integrating an automated price update mechanism, the platform ensures that every transaction is accurate and reflects the latest market value.

Customers can interact with the platform by selecting their preferred gold karat, browsing specific categories, and completing their purchases online. Meanwhile, administrators have full control over the system to monitor daily orders, track total sales, and update the inventory. The admin panel provides essential statistics and allows the owner to manage product details, including weight and workmanship fees, ensuring the business remains competitive and efficient.

2. Project Scope:

The scope of the project includes user management with role-based access, where admins can fully control product management (adding, updating, and deleting jewelry items) and inventory tracking. A key feature of the scope is the Dynamic Pricing Engine. Customers can view products, filter by gold purity, and make purchases securely. The system will be developed using “JAVA” for the backend logic, “MySQL” for the database system, and “JAVA-FX Scene Builder” for the user interface.

3. System & Database Requirements:

To ensure the success of the Dubai Jewelry platform, the following functional and data requirements must be met:

1. Dynamic Price Calculation: The system must automatically calculate the final price of any jewelry item by combining its weight, the current karat rate, and specific workmanship fees.
2. Hierarchical Navigation: The database must support a structured browsing flow, allowing customers to filter products first by gold purity (Karat) and then by jewelry category (e.g., Rings, Necklaces).
3. Inventory & Warehouse Management: The system must track stock levels across different branches or warehouses and alert the administrator when quantities are low.
4. Secure Transaction Logging: Every order must be recorded with a "Price_at_Time" snapshot to ensure that even if gold prices change later, the customer's original transaction record remains accurate.
5. Role-Based Access Control: The system must distinguish between "Customers" (who browse and buy) and "Staff /Admins" (who manage products, view sales reports, and update inventory).
6. Comprehensive Sales Reporting: The database should provide the ability to generate financial reports based on daily or monthly sales, as well as sales grouped by gold karat or category

4. Client Information:

We have chosen Dubai Jewelry as our client, as they can provide valuable assistance in defining the project requirements and supplying the necessary data for development. The proposed system aims to extend the store's existing services to include an online platform, enabling customers to browse and purchase jewelry with real-time pricing based on gold market rates. Dubai Jewelry will serve as our primary reference and partner institution throughout the design and implementation phases, ensuring that the system meets real-world business needs and supports the store's mission of providing high-quality jewelry with price transparency.

Contact Information :

- Name: Dubai Jewelry.ps
- Address: EL Ersal st, City Mall, Ramallah
- Phone: +972566554432
- Email: info@dubaijewelry.ps

5. Entities description:

The Karat table defines the gold purity levels available in the store, such as 21K, 18K, and 14K. It stores the Karat ID, name, and the current price rate per gram.

The Category table organizes jewelry into specific types, such as rings, necklaces, or bracelets. It contains the category ID and the category name, helping customers navigate the catalog efficiently.

The Product table holds general information about every jewelry item available for sale. It includes the product ID, name, weight, and stock quantity. Each product is linked to a specific Karat and Category to determine its base value and classification.

The Product Details (Workmanship) stores the labor cost or "مصنعية" associated with each specific piece. This value, combined with the weight and current Karat rate, allows the system to calculate the final price displayed to the customer.

The User table stores information about people interacting with the system, including the user ID, name, address, phone number, and account credentials. It also distinguishes between regular customers and administrators (Staff).

The Orders table records customer transactions, including the unique order ID, the date of purchase, and the total price calculated at the time of the transaction. It is linked to the user who placed the order and specifies the payment method used.

The Order List relation (between Orders and Product) represents the specific items included in each order. It stores the list ID, the amount (quantity) of each product purchased, and the specific price of the item at the exact moment the order was placed to lock in the gold rate.

The Payment Method table lists the various ways customers can pay for their jewelry, such as Cash on Delivery or Credit Card, providing flexibility during the checkout process.

6. Relationships:

1. **Customer-Orders (1-M):** Each customer can place many orders, while each order is associated with exactly one customer.

2. **Staff-Orders (1-M):** Each staff member (Admin) can manage and process many orders, while each order is managed by at most one staff member.

3. **Orders-Product (M-M):** Each order can contain many products, and each product can appear in many orders. (Association attributes: Amount, Price_at_Time in the Order_List relation).

4. **Karat-Product (1-M):** Each Karat purity level (e.g., 21K, 18K) can be assigned to many products, while each product belongs to exactly one Karat level.

5. **Category-Product (1-M):** Each category (e.g., Rings, Bracelets) can contain many products, while each product is classified under exactly one category.

6. **Payment Method-Orders (1-M):** Each payment method can be used in many orders, while each order uses exactly one payment method (e.g., Cash or Card).

7. Queries:

1. Retrieve all gold purity levels stored in the system, including the karat name, price per gram, and last updated date.

2. Retrieve all jewelry categories (e.g., bracelets, necklaces, rings) available for a selected gold karat.

3. Retrieve all products stored in the database, including product name, product weight, stock quantity, karat type, and category.

4. Retrieve basic product information, such as product name and product weight, to provide a simple overview of the available jewelry items.

5. Retrieve all available payment methods that customers can use when placing an order, such as Cash on Delivery, Credit Card.

6. Retrieve all system users with their basic information, including username, first name, last name, email, phone number, and user role.

7. Retrieve all customer records stored in the Customer table, including the customer user ID and registration date.

8. Retrieve all admin records stored in the Admin table, including the admin user ID and join date.

9. Retrieve all orders stored in the database, including customer ID, admin ID, payment method ID, order date, and total price.

10. Retrieve all order list records to show the relationship between orders and products, including product quantity and the price stored at the time of purchase.

11. Retrieve all products with their related karat and category information, allowing the system to display each product with its gold purity level and jewelry type.

12. Retrieve the full product price details, including product name, karat name, category name, product weight, price per gram, workmanship fee, and the calculated final price.

13. Retrieve all orders with customer and payment method information, including customer first name, customer last name, selected payment method, order date, and total price.

14. Retrieve detailed information for a specific order, including the products included in the order, quantity, price at the time of purchase, and subtotal for each product.

15. Retrieve the best-selling products based on the total quantity sold across all orders.

16. Retrieve total sales grouped by jewelry category to identify which product types, such as rings, necklaces, bracelets, or earrings, generate the highest sales.

17. Retrieve daily sales reports, including the order date, number of orders, and total sales revenue for each day.

18. Retrieve monthly sales reports, including the sales year, sales month, number of orders, and total monthly revenue.

19. Retrieve customer order information with contact and address details, including customer name, phone number, city, street name, order date, total price, and payment method.

20. Retrieve the calculated final price for each product based on product weight, current karat price per gram, and category workmanship fee.

8. ER Diagram:

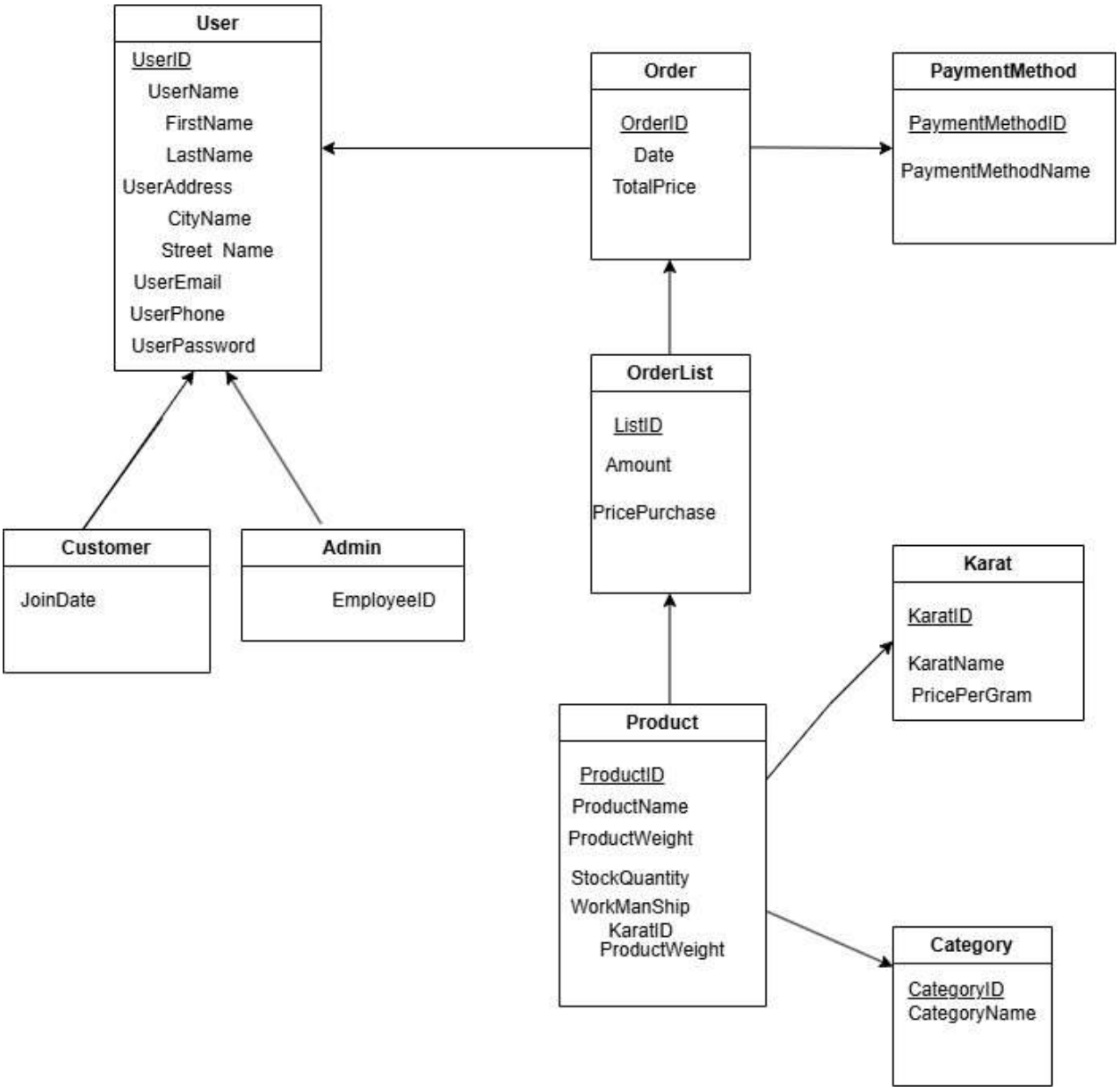


Figure 1 : ER Diagram

As shown in the provided ER diagram, the data model connects the system's main entities: User (with its specializations Customer and Admin), Order, OrderList, Product, Karat, Category, and PaymentMethod through a set of relationships that describe the flow of sales, product management, and user interactions. These relationships make it possible to trace every product from its categorical classification and pricing details (based on gold purity) to its inclusion in specific customer orders. By linking entities using primary and foreign keys and recording

critical details such as product weight, workmanship , and total prices, the database supports accurate inventory management, precise pricing calculations, and reliable transaction tracking.

At the core of the user management side, the User entity acts as a general base for all individuals interacting with the system. It is linked to Customer and Admin through a specialization relationship. This design allows the system to store common attributes like names, addresses, and contact info in the User entity, while maintaining specific data such as "JoinDate" for customers and "EmployeeID" for administrators. The User (specifically as a Customer) is connected to the Order entity, representing that each order is placed by a specific user, which is essential for tracking purchase history and managing customer accounts.

On the sales and transaction side, the Order entity serves as the primary record for a completed purchase, storing the date and total price. It is connected to the PaymentMethod entity, which defines how the transaction was settled (e.g., Cash, Card). To handle the details of each purchase, the Order is linked to OrderList (the junction or line-item entity). This relationship allows a single order to contain multiple items, recording the specific "Amount" (quantity) and "PricePurchase" for each entry, ensuring that historical price data is preserved even if product prices change in the future.

The Product entity is the central hub for item information, linked to OrderList to show which products are being sold. Each product is characterized by its weight (ProductWight), stock quantity, and "WorkManShip" (labor cost). To ensure accurate pricing and organization, Product is connected to two descriptive entities: Category and Karat. The link to Category allows for grouping products (e.g., Rings, Necklaces), while the link to Karat is vital for the jewelry context; it connects the product to specific gold purity levels and their current "PricePerGram." This structure ensures that the system can dynamically calculate product values based on weight and the fluctuating gold market. Overall, these relationships create a consistent view of the business workflow, linking user roles, detailed product specifications, and structured order processing into one integrated database.

9. Normalization:

The Dubai Jewelry database design was normalized to reduce redundancy, avoid update anomalies, and make sure that each table represents one clear concept. The system includes several main entities such as User, Customer, Admin, Product, Karat, Category, Order, OrderList, and PaymentMethod. By separating these entities correctly, the database becomes easier to manage and more consistent. For example, product information is stored in the Product table, gold purity information is stored in the Karat table, jewelry type information is stored in the Category table, and payment information is stored separately in the PaymentMethod table. This separation helps ensure that each non-key attribute depends only on the key of its own table.

First Normal Form (1NF) was achieved by making sure that all attributes are atomic and that there are no repeating groups inside a single field. For example, the User table stores user information such as name, email, phone number, city, street, and password as separate attributes instead of storing multiple pieces of information in one attribute. Also, an order may contain many products, but these products are not stored as a list inside the Order table. Instead, the OrderList table is used to represent each product included in an order as a separate row. This means that each record in OrderList contains one product, its quantity, and the purchase price at the time of ordering. This design satisfies 1NF because each cell contains a single value and repeated product lists are removed from the Order table.

Second Normal Form (2NF) was achieved by removing partial dependencies, especially in tables that represent relationships between two entities. The main example is the OrderList table, which connects Orders and Products. The information in this table, such as Amount and PricePurchase, depends on both the OrderID and the ProductID together, not on only one of them. The quantity of a product is not a property of the product alone because the same product can be purchased in different quantities in different orders. It is also not a property of the order alone because one order can contain many different products. Therefore, placing Amount and PricePurchase in the OrderList table prevents partial dependency and keeps the design in 2NF.

Third Normal Form (3NF) was achieved by removing transitive dependencies and separating descriptive data into independent tables. For

example, the Product table does not store the full karat details or category details directly. Instead, it references the Karat table and the Category table using foreign keys. This prevents repeating the same karat price or category name for many products. If the price per gram for a karat changes, it only needs to be updated in the Karat table instead of updating many product records. Similarly, the Order table does not store the payment method name directly. Instead, it references the PaymentMethod table, which avoids repeating payment method descriptions in every order. This reduces redundancy and prevents update anomalies.

The design also handles dynamic pricing correctly by storing the current gold price in the Karat table while keeping the purchase price at the time of the order in the OrderList table. This is important because gold prices change over time, but old orders must keep the exact price that was used when the customer made the purchase. Therefore, PricePurchase is stored as historical transaction data in OrderList, while PricePerGram remains in the Karat table as current pricing data. This separation prevents inconsistencies between current product prices and old order records.

Overall, the Dubai Jewelry database is normalized up to Third Normal Form. The design removes repeating groups, avoids partial dependencies in relationship tables, and separates lookup or descriptive entities such as Karat, Category, and PaymentMethod from transaction entities such as Orders and OrderList. As a result, each table represents one main concept, and each non-key attribute depends only on the primary key of its table. This improves data consistency, reduces redundancy, and supports accurate product management, dynamic pricing, inventory tracking, and order processing.

10. References:

[1]: Dubai Jewelry – مـجـوهـرات دبـي. Official social media page, used for client/business information and contact details. Accessed: June 2026.